

BHASKAR MANDAL

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LinkedIn

GitHub

Portfolio

Education

Indian Institute of Technology, Kharagpur

M.Tech in Computer Science And Data Processing

2025 – 2027

CGPA: 7.33/10

Experience

Technology Research Intern | IIT Kharagpur

June 2026 – July 2026

- Developed optimized Python solutions for computational research, improving processing efficiency by 15% - 20%.
- Designed and optimized algorithms, reducing computational runtime by 20% - 35% through efficient logic.

Data Science Intern | Internship Studio | [Certificate](#)

Jan 2026 – June 2026

- Built a Credit Card Eligibility Prediction System using **Random Forest** classification, achieving **97.5%** Prediction accuracy. [Click](#)
- Developed a Future Sales Prediction Model using **Random Forest** regressors, yielding **94.8%** accuracy. [Click](#)

Data Analyst Intern | KultureHire | [Certificate](#)

Dec 2025 – Jan 2026

- Leveraged **SQL** and **EDA** to analyze **98K+ Gen-Z career survey records**, reducing data processing time by approximately **35%** through optimized SQL queries while uncovering actionable business insights using trend, correlation, and visualization analysis.

Projects

Hybrid RAG Based Research Assistant | *Python, Streamlit, Gemini LLM, ChromaDB*

June 2026

[GitHub: Repo Link](#)

[Live Demo: Streamlit Link](#)

- Developed a **Retrieval-Augmented Generation (RAG)** web application to answer questions from uploaded research papers using **Google Gemini API** and **ChromaDB**.
- Implemented document ingestion for **PDF, DOCX, TXT, and Markdown** files with **text chunking, vector embeddings, and semantic search**.
- Generated **context-aware, citation-based** responses by retrieving relevant document chunks through a **vector database**.

Multi-Agent Based Cooking System | *Python, Streamlit, Gemini LLM, Agents*

July 2026

[GitHub: Repo Link](#)

[Live Demo: Streamlit Link](#)

- Developed a **multi-agent AI cooking assistant** for recipe recommendation, ingredient substitution, nutrition analysis, and meal planning.
- Designed a **multi-agent workflow** where specialized AI agents collaborate to generate accurate, context-aware responses using the **Google Gemini API**.
- Implemented modular Agent **Orchestration** to improve scalability, maintainability, and response quality.

American Sign Language Recognition System | *Python, MediaPipe, Scikit-learn, OpenCV*

July 2026

[GitHub: Repo Link](#)

[Live Demo: Streamlit Link](#)

- Developed a real-time **ASL** recognition system using **MediaPipe** landmark detection, achieving **98.73%** accuracy.
- Designed an end-to-end **computer vision** pipeline for landmark extraction, feature engineering, gesture prediction, improving inference speed by approximately **30%**.

AgroCare: Plant Disease Detection and Treatment System | *Python, TensorFlow, CNN*

July 2026

[GitHub: Repo Link](#)

[Live Demo: Streamlit Link](#)

- Developed a **CNN-based plant disease detection and treatment recommendation system**, achieving **98.12%** accuracy, **98.08%** precision, **98.05%** recall, and **98.06%** **F1-score** across **38** plant disease classes.
- Designed an end-to-end **deep learning pipeline** for image preprocessing, CNN model training, evaluation, and real-time inference, reducing inference latency by approximately **30%**.

Technical Skills

- Programming Languages:** Python, C++, Dart.
- Databases:** MySQL, SQL, RDBMS.
- Core Computer Science:** DSA, OOP, DBMS.
- Generative AI:** LLMs, Prompt Engineering, Multi-Agent Systems, Agent Orchestrators, LangChain, LangGraph.

Certifications

- Prompting Essentials - *Google* [Click](#)
- AI Essentials - *Google* [Click](#)
- SQL - *Udemy* [Click](#)
- Power BI - *Skill Course* [Click](#)